MIL-A-18218B
19 November 1984
SUPERSEDING
MIL-A-18218A
4 May 1979

MILITARY SPECIFICATION

ANALYZERS, FLUE-GAS, ORSAT-TYPE, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable Orsat-type analyzers for the measurement of the volumetric percentage of CO_2, O_2, and CO occurring in flue-gases.

1.2 Classification. Flue-gas analyzers shall be of the following sizes, as specified (see 6.2.1):

Size 1 - 100 milliters (ml).
Size 2 - 50 ml.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The analyzer shall be a metal-frame-mounted, portable assembly, consisting of a jacketed burette, manifold with control valves, absorption pipettes with integral or close-coupled reservoirs for absorption chemicals, an expansion bag(s) or chamber(s), a leveling bottle, a dirt-excluding protective enclosure, necessary connectors, flexible tubing and stoppers, and when specifed (see 6.2.1), an aspirating bulb, sampling line, filter, and absorption chemicals.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish an analyzer for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The analyzer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in

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the analyzer being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Design and performance.

3.5.1 General. The analyzer shall be designed to withstand rough usage and for easy and trouble-free manipulation, rapid and thorough gas absorption, easy readability, and portability as a unit. Jars and bumps, such as those encountered during transportation and under the conditions of intended use, shall not cause breakage, misalinement, loosening of components, or leaks.

* 3.5.2 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5.3 Arrangements. The burette and pipettes shall be arranged in a row, from right to left, as follows: Measuring burette, CO_2 pipette, O_2 pipette, and CO pipette. The bottom of the frame, or the support points, shall touch a flat supporting surface simultaneously. With the analyzer standing on a flat surface, the long axis of each burette and pipette shall be essentially vertical. The arrangement shall be such that rubber connectors or other components capable of absorbing gases from flue-gas shall present as small an area for contact with flue-gas samples as possible.

3.5.4 Accuracy. Each reading shall indicate the percentage by volume of gas absorbed by the absorbing solution from the measured sample during analysis within the following limits:

Size 1 - $\pm$ 0.2 percent of burette volume.
Size 2 - $\pm$ 0.3 percent of burette volume.

3.5.5 Accessibility. Each component shall be readily demountable with no damage to the component or adjacent components during removal and replacement.

3.5.6 Leakage. There shall be no gas leakage through closed valves or between the analyzer and the atmosphere. Chemicals shall not leak or creep past or through seals, connectors, valves, or cocks.

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3.6 Burette, jacket, and scale. The burette and jacket shall be of clear, high grade materials free of striae, whorls, or other imperfections which would result in false indications of liquid levels or scale reading. The materials shall not become fogged or discolored with age. The burette shall be of glass thoroughly annealed after being formed. The jacket shall be of clear glass or of clear plastic. Changes in cross-sectional area in the burette shall be made with generously rounded curves. Wall thickness shall be substantially uniform in each portion, and changes in wall thickness between different-diameter portions shall be uniform. On or adjacent to the lower portion of the burette, there shall be a scale marked with parallel division lines from zero to 21 (or 20.8 percent) at 0.1 percent intervals for size 1 and 0.2 percent intervals for size 2, and shall be marked with numerals at 1 percent intervals for size 1 and at 2 percent intervals for size 2, starting at zero and ending at 20. The division lines shall be sharply defined and shall be of a deep dark color. The shortest division lines shall be not less than 5/32 inches long. The lines at integral points shall be somewhat longer that the intermediary lines. The length of the scale shall be not less than 6-1/2 inches for size 1 not less than 4 inches for size 2 analyzers. Scales shall be etched and filled in with a fusible material on the burette or shall be marked in enamel on a corrosion-resistant metal strip. The latter, when used, shall be assembled with the burette in such a way that the whole length of scale will remain in contact with the burette, and so that the scale will not move with respect to the burette because of vibration or other movement of the analyzer, or because of changes in dimension due to temperature. Scale markings and numerals shall not chip, peel, or fade with continued use.

3.7 Manifold and valves. The manifold and valves shall be of nonbrittle, nonfragile materials which shall not be affected chemically and shall not deteriorate physically as a result of contact with air or flue-gas, or with the liquids used in the analyzer. Glass manifolds and glass stopcocks will not be acceptable. Means shall be provided for easy cleaning of gas passages without extensive disassembly. Valve control knobs and handles shall be designed for easy and interference-free manipulation. Valves shall close gas-tight with light seating-pressure and shall not stick or bind in normal use. Valves shall be capable of being opened and fully closed 10,000 times without showing leakage as a result thereof.

3.8 Absorption pipettes and reservoirs. Absorption pipettes may be integral with or separable from reservoirs. When they are separable, assembly shall be effected by means of a compression-type fitting, which shall provide a leak-tight joint without breaking the pipette. The absorption space shall be filled to the neck of the pipette capillary with contact-extending material, or shall be provided with a bubbler, having at least four bubbling openings, approximately 1 mm in diameter, near the bottom of the pipette and equipped with an automatic valve. Contact-extending material shall be so arranged that jostling and jarring during transport and handling will not cause the material to pack or fall through the bottom of the pipette or otherwise to become dislodged. Bubbler pipettes shall be so constructed that all gas will have to pass through the absorbing solution and during the absorbing operation no liquid, which could be drawn into the manifold if the flow direction were reversed, shall be trapped by the valve. The valves of bubbler pipettes shall easily unseat when gases are withdrawn to the burette. It shall not be necessary to tap any pipette to release the last traces of free gas. Edges on the ends of pipettes shall be made smooth. In the upper

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part of each reservoir there shall be suitable means for the combined purpose of connecting the reservoir with an expansion bag or chamber by means of flexible rubber or synthetic rubber tubing, and for filling with absorbing solution. When the connection between the reservoir and the expansion bag or chamber is not of permanent nature, then a suitable means shall be provided to make the reservoir opening airtight and leakproof when the analyzer is out of use. Glass parts shall be fully annealed and all parts shall be free of any tendency to warp, become embrittled, crack, or otherwise deteriorate with normal aging. Pipette and reservoir shall be readily removable as a unit.

3.9 Expansion bag or chamber. The expansion bag(s) or chamber(s) shall have sufficient capacity to accept the gas displaced by the liquid-rise in the reservoir during analysis, without the development of a back-pressure exceeding 1 inch of water. They shall be leakproof and shall not deteriorate rapidly due to contact with air or the gases released by the absorbing solution.

3.10 Leveling bottle. The leveling bottle shall be made of a transparent material which shall not become fogged or discolored with age. Provided, at or near the bottom, shall be a narrowed extension or neck, for connection to flexible rubber or synthetic rubber tubing. The extension or neck shall be so formed that the connection tubing will not easily work loose due to normal manipulation.

3.11 Protective enclosure. The protective enclosure shall be of rigid material and construction. It shall provide a snug closure so that dirt will be excluded. Positive means shall be provided to insure that the analyzer and protective enclosure will not separate when the assembly is lifted by its upper end. The enclosure shall be easily removable. When the enclosure is removed, the apparatus shall be completely open so all parts can be seen.

3.12 Connectors, flexible tubings, and stoppers. Connectors, flexible tubing, and stoppers shall be of the highest grade material and manufacture available for the intended use. They shall not check, or become embrittled or swollen and pasty with normal aging or due to contact with the operating liquids and gas samples with which they normally come in contact.

3.13 Aspirating bulb. When specified (see 3.1 and 6.2.1), the aspirating bulb shall be equipped with leakproof, nonsticking valves. It shall be quick-acting and shall be easily workable with one hand.

3.14 Sampling line. When specified (see 3.1 and 6.2.1), the sampling line shall be of rubber, or synthetic rubber tubing of the length specified.

3.15 Filter. When specified (see 3.1 and 6.2.1), the filter shall effectively prevent the entrance of soot, fly ash, and water condensate into the analyzer without the introduction of appreciable resistance to the flow of gas. It shall be readily cleanable.

3.16 Absorption chemicals. When specified (see 3.1 and 6.2.1), absorption chemicals shall be supplied in the quantities specified. They shall have high absorption capacity and shall not deteriorate in storage. Powder and the necessary acid for the CO absorption solution shall be mixed, unless otherwise

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specified (see 6.2.1). Safe handling procedures and material safety data sheets for absorption chemical shall be supplied by each absorption chemical manufacturer as required by DoDINST 6050.5.

3.17 Treatment and painting. Unless otherwise specified (see 6.2.1), the equipment shall be treated and painted in accordance with the manufacturer's best standard practice.

3.18 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The nomenclature of the item shall be "ANALYZER, FLUE-GAS, ORSAT-TYPE, PORTABLE."

3.19 Repair or maintenance parts and accessories. Repair or maintenance parts and accessories shall be furnished as specified (see 6.2.1).

3.20 Workmanship. The quality of workmanship shall be such as to produce analyzers that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the unit.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one analyzer when a first article is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

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* 4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one analyzer. All analyzers of the same size offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

* 4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

* 4.3.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 4.0 percent defective.

* 4.4 Examination. Each analyzer shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.5 Tests. Each unit selected shall be tested, and any unit failing to pass the following tests, as applicable, shall be rejected. A certificate of compliance from an independent laboratory, approved by the contracting officer, will be accepted as evidence that the unit has passed the required test.

4.5.1 Leakage test. Each assembled analyzer shall be tested for leakage of air into, or out of, the analyzer with a pressure of approximately 0.5 pounds per square inch below and above atmospheric, respectively. The test shall be so conducted that every point at which leakage could occur will be subjected to the test pressure. The analyzer shall show no leakage when the pressure has been applied for 10 minutes.

4.5.2 Accuracy test. Each analyzer, assembled and filled to the normal levels with the appropriate operating liquids, shall be tested for accuracy, using a gas sample composed of a mixture of 8 $\pm$ percent CO_2, 4 $\pm$ .01 percent CO, 4 $\pm$ .01 percent O_2, and the remainder nitrogen. Absorption shall continue until two consecutive passages of the sample, into and out of each absorption pipette, result in no further change in volume of the sample. Before analysis, the analyzer shall be flushed once, with a gas mixture of the composition specified for the accuracy test.

* 4.6 Packaging inspection. The preservation, packing, and marking of the analyzer shall be inspected to verify conformance to the requirements of section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2.1).

* 5.1.1 Level A. Each complete analyzer shall be packaged, method III in accordance with MIL-P-116, in a close-fitting box conforming to PPP-B-636, class weather-resistant. The components of the analyzer shall be cushioned to prevent movement and damage. The box shall be closed in accordance with the appendix to the box specification, method V.

* 5.1.2 Level C. Material shall be packaged in accordance with ASTM D3951.

* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

* 5.2.1 Levels A, B, and C. Packing shall be accordance with table I of MIL-STD-794 for the applicable level specified. Containers shall be selected from table I for the appropriate level. Open crates shall not be used for Level A or B packing.

* 5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The analyzers covered by this specification are intended for analysis and measurement of combustion products of fuels (flue-gas).

6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Whether aspirating bulb is required (see 3.1 and 3.13).
d. Whether sampling line is required, and length required (see 3.1 and 3.14).
e. Whether a filter is required (see 3.1 and 3.15).
f. Whether absorption solution powder and quantities required (see 3.1 and 3.16), and whether the CO absorption solution powder and acid shall be supplied in other than the mixed form.
g. When first article is required for inspection and approved (see 3.2, 4.2.1, and 6.4).
h. Treatment and painting if other than specified (see 3.17).
i. Applicable repair or maintenance parts, and accessories (see 3.19).
j. Level of packaging and level of packing required (see 5.1 and 5.2).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be

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delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
4.6	Certificate of compliance	UDI-E-24007	

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

* 6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one, unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Navy - YD
Air Force - 99

User:

Air Force - 82

Preparing Activity:

Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-18218B	ANALYZERS, FLUE-GAS, ORSAT-TYPE, PORTABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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